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Scalable machine learning for urban thermal morphology classification: clustering high-resolution microclimate and dynamic mapping of radiant temperature

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Abstract

Recent advances in ultra-fine and ultra-fast urban climate modelling have improved the identification and representation of complex urban environments. However, their practical application of Mean Radiant Temperature (MRT) time series generated by these tools remain difficult to interpret for non-expert stakeholders, necessitating a transduction of complex numerical data into actionable spatial information. Leveraging LUCIDIT—an ultra-fast simulation tool enabling high-resolution MRT maps across continuous urban domain—this research proposes a scalable, automated framework for classifying different “thermal morphologies” based on the MRT time series analysis. The capacity of unsupervised machine learning to capture the nonlinear relationships of urban morphology is investigated through a rigorous comparison between a physics-based heuristic approach, a shape-based algorithm (K-Shape), and K-Means clustering applied to both engineered thermal features and MRT time series. Results demonstrate that applying K-Means directly to the raw time series outperforms the other methods in terms of both computational efficiency and mathematical robustness, as confirmed by the Silhouette Score and Davies-Bouldin Index. Furthermore, the physical accuracy and interpretability of the data-driven clusters are successfully validated by projecting the spatial outputs into a novel 2D Thermal Phase Space. This computationally efficient methodology, applicable at different urban scales, converts the physical complex and synergistic dynamics of the urban microclimate into intuitive and easy-to-use spatial thermal maps. It connects advanced numerical simulations to practical urban design, providing a compact representation of urban thermal behaviour and enables the large-scale identification of recurrent microclimatic patterns fundamental for urban heat mitigation strategies.

Keywords Urban thermal classification, Machine Learning, High-resolution microclimate clustering, Urban sustainability, Resilience urban planning, Urban Mean Radiant Temperature

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1 Introduction

Urban sprawl, hyper urbanism and connected anthropogenic impacts, combined with climate change, Urban Heat Islands (UHIs), and global warming highlight the importance of monitoring urban climate evolution and evaluating the effectiveness of existing plans and planning practices for resilient and sustainable cities (Ali et al., 2026; Alrababaa et al., 2025; Choi, 2025; Intergovernmental Panel on Climate Change (IPCC), 2023; Putra & Phinn, 2026; Zhao et al., 2021).

Human thermal comfort in urban outdoor environments is primarily governed by the Mean Radiant Temperature (MRT), a crucial parameter that synthesizes the complex shortwave and longwave radiative exchanges between the human body and the surrounding urban morphology (Johansson et al., 2014; Mayer & Höppe, 1987). Indeed, MRT is the most important environmental parameter for the thermo-physiological indices assessment (Pierucci et al., 2025), i.e. the Universal Thermal Climate Index (Bröde et al., 2012), the Physiological Environmental Temperature (PET) (Höppe, 1999), and the outdoor Predicted Mean Vote (Fanger, 1970). It directly connects variable microclimatic conditions to human thermal perception and providing useful information on health risks (Di Napoli et al., 2018; Muthers et al., 2010; Pantavou et al., 2025). Consequently, high-resolution spatial and temporal mapping of MRT is essential to accurately identify urban thermal hotspots and predicting UHI formation (Lin et al., 2017; Mayer & Höppe, 1987).

Traditionally, urban microclimate assessments have relied on computationally intensive physical models e.g., standard Computational Fluid Dynamic (CFD) tools or ENVI-met (*Ansys | Engineering Simulation Software*, n.d.; *ENVI-Met*, n.d.; *The COMSOL® Software Product Suite*, n.d.). The CFD models allow for an explicit representation of the air velocity and temperature fields, but require significant computational effort and prohibitive computation times, especially for high resolution city-scale simulation purposes. To overcome these obstacles, studies have increasingly adopted decoupled radiative models, such as SOLWEIG (Brozovsky et al., 2022; Chen et al., 2016; Kastner & Dogan, 2022; Lindberg et al., 2008; Perini et al., 2017; Rashtian et al., 2025). But these models heavily rely on intensive ray-tracing algorithms and explicit view factor calculations, and remain computationally expensive for generating continuous, high-resolution, multi-day simulations over entire metropolitan areas. Starting from a recent research, our present study uses LUCIDiT (Lean Urban Comfort Intelligent Digital Twin) (Baia et al., 2026), an ultrafast modelling approach that decouples these intensive processes, enabling the rapid generation of spatial high-resolution MRT maps over continuous multi-day periods.

These advanced simulation tools solve the computational bottleneck but introduce a critical challenge due to data interpretability. High-resolution urban simulations generate massive space–time data series encompassing millions of individual pixels. Translating this “big data” into actionable knowledge is a recognized challenge in various fields.

Recent literature highlights that space–time series representation, coupled with data mining and machine learning (ML) techniques, is highly effective for managing complex dynamics in transportation (Hu et al., 2018), healthcare (Xi et al., 2018), seismology (Morales-Esteban et al., 2014), and climate science (Y. Liu et al., 2015). Specifically, space–time series clustering has attracted significant interest for grouping heterogeneous temporal trajectories into meaningful geographic patterns (Belhadi et al., 2020). Some authors have shown how digital signal decomposition of nonlinear and non-stationary times series allows the identification in long-term variations induced by any urban-morphological effects connected to multiple clusters (H. Liu et al., 2025).

On the other hand, in recent studies the development of sustainable urban areas for climate change mitigation and adaptation is supported by ML analyses. Some examples involve the forecasting of parametric urban morphogenesis with Convolutional Long Short-Term Memory network (Alrababaa et al., 2025), and Geographic Information Systems (GIS)-based views-heed analysis (Das et al., 2025), the combined use of remote sensing, GIS data and artificial neural networks (Fontana et al., 2023), as well as thermal energy simulations at different urban and neighbourhood scales (Mistry & Mehrotra, 2025).

To reduce computation times and in the absence of detailed data for individual buildings, integrated modelling methods have been applied, combining remote sensing, geospatial and statistical analysis, and ML techniques, to characterize complex urban systems and study their interaction with the climate at large and hyperlocal scales, and with landforms (e.g. combinations of densely built-up, green areas and/or Nature-based Solutions, shadows diversity) in the perspective of climate responsive and sustainability resilience urban planning (Ali et al., 2026).

Building clustering approaches that leverage available local information, geospatial data and multivariate time series clustering can be used to generate a customized set of archetypal buildings, forecasting building energy analyses and different visualization methods for urban heat energy planning (Gianniou et al., 2018; Sanhudo et al., 2021; Torabi Moghadam et al., 2019; Yerbury et al., 2025).

Some recent works have provided building clustering approaches as an alternative to clustering based on energy performance indicators (De Jaeger et al., 2020; Iglesias & Kastner, 2013; Laurinec et al., 2016) and others

have suggested the use of K-means method to increment accuracy in forecasting of adaptive building power load (Brentan et al., 2018; Majidi, 2023).

Despite these advances in energy and spatial clustering, handling microclimatic MRT data requires a specific paradigm. For non-expert stakeholders, such as urban planners and policymakers, raw multidimensional MRT datasets are too dense to be directly usable. While established commercial suites and graphic engines (such as ENVI-met Leonardo) provide robust platforms for visualising these outputs through interactive hourly snapshots or animations, they function strictly as visual post-processors rather than analytical data-mining engines. Interpreting a multi-day microclimatic event through these conventional tools requires that users manually scan through dozens of individual hourly maps (e.g., 96 separate frames for a 4-day simulation). This fact implies a process that is highly time-consuming and susceptible to cognitive overload. Using traditional data reduction techniques—such as mapping daily temperature averages, absolute maximums—fails to capture the fundamental temporal dynamics, such as directional solar exposure, that produces human thermal stress. Many literature studies on urban microclimate, attempts to condense space–time data range from straightforward risk indicators, such as mapping the cumulative number of hours exceeding a specific discomfort threshold, to more sophisticated statistical indices. However, these methods still fail to capture the temporal signatures embedded within the urban fabric.

As highlighted in a recent comprehensive review (Marey et al., 2025), the most current ML applications in urban microclimate analysis operates under supervised, regression-based paradigms. These frameworks typically map a priori geometric features (e.g., Local Climate Zones) to thermal outputs to bypass physical simulations (AlKhaled et al., 2024; Briegel et al., 2024; Ketterer & Matzarakis, 2016; Lau et al., 2023; Shi et al., 2025). The proposed regression predictive models work as black boxes that map inputs to outputs without extracting the underlying spatiotemporal archetype that drives the urban system. Furthermore, the static morphology proxies utilized in these literature studies, such as Sky View Factor (SVF), generally suffer from structural orientation blindness, mixing different solar exposure regimes. Alternative unsupervised spatial clustering approaches often rely on 2D Land Surface Temperature (LST) maps derived from satellite imagery (Feng et al., 2025; Jiang et al., 2026; Li et al., 2025; Manyanya et al., 2024). However, the low spatial and temporal resolution of satellite data is often insufficient for this purpose, failing to capture the three-dimensional details and complexity of urban areas.

Only very recently, unsupervised data-mining techniques has been applied to directly characterize urban

microclimatic zone from simulated time-series datasets. A recent important work (Nevers et al., 2025) has demonstrated that clustering multi-temporal comfort index, such as CFD-derived data, can isolate distinct microclimatic behaviour at a neighbourhood scale.

Starting from this background, with the aim to match advanced numerical simulations and applied urban planning, our research employs an unsupervised clustering directly on high-resolution, multi-day MRT time series to predict temperature from fixed geometry. To achieve this goal, the following research questions are addressed:

1. Which clustering architecture provides the most robust and mathematically consistent classification of high-resolution MRT time series within a complex urban environment?
2. Can unsupervised clustering be applied directly to continuous MRT trajectory identifying physically meaningful urban thermal archetype, that reflects different thermodynamic behaviours and different solar exposure regimes?
3. How the resulting thermal archetypes can be translated into an interpretable and operational framework that supports UHI mitigation and climate adaptation strategies?

To answer these questions our present research, provide a multi-layered scientific approach. First, by coupling the unsupervised ML pipeline with the ultrafast LUCIDiT model (Baia et al., 2026), the applicability of the proposed framework is demonstrated over a large heterogeneous urban domain (built-up areas with and without greenery), while maintaining a very low computational costs. Second, a rigorous methodological comparison is carried out: a physics-based heuristic approach is set as a baseline and benchmarked against a shape-matching algorithm (K-Shape) (Paparrizos & Gravano, 2016) and a variance-minimization algorithm (K-Means) (Ikotun et al., 2023; MacQueen, 1967). The latter is applied to both engineered scalar features and raw continuous MRT data series. By validating the clustering results across a novel 2D thermal phase space and comparing them with urban morphological factors, our research work demonstrates how fundamentally data-driven algorithms can learn and evolve the thermodynamics of any urban system. Finally, to solve the systemic limitation of the absolute magnitude loss in unsupervised clustering, a novel dual-layer visualization framework is introduced. A risk-prioritized diagnostic tool for targeted urban heat mitigation and climate change adaption is realized using alpha channel opacity modulation based on absolute thermal threshold exceedance.

The paper is organized as follows: Sect. 2 describes the study area and data preparation, Sect. 3 details

the methodology, including the physical baseline, the selected unsupervised ML approaches, the feature engineering process and the clustering validation. Section 4 summarizes the main findings, e.g. the discussion of the extracted thermal urban archetypes, its interpretation and their spatial translation for urban planning. Section 5 outlines concluding remarks.

2 Data source

2.1 The study area

The geographical domain selected for this study encompasses a large district in the North-Western sector of Florence, Italy (bounding box coordinates: Longitude 11°13'26.4" to 11° 15' 18" E, Latitude 43°47'16.8" to 43°48'28.8" N). Covering an expanded area of approximately 5.5 km², this specific urban fraction was chosen to rigorously test the scalability of the clustering framework across a highly heterogeneous morphological matrix (Fig. 1).

The domain encompasses a complex mix of narrow historical street canyons, wide open squares, dense residential blocks, industrial facilities, and different greenery. To enable high-resolution microclimatic simulations at the district scale, this physical domain was subsequently translated into a digital representation. The specific methodological details regarding the discretization and generation of this digital urban dataset are discussed in Sect. 2.2.

From a climatological and hyperlocal microclimatic point of view, Florence occupies a transitional zone between a humid subtropical (Cfa) and a hot-summer Mediterranean (Csa) climate (Beck et al., 2018; Kottek et al., 2006). The region is usually characterized by temperate, humid winters and hot, dry, and often sultry summers frequently marked by severe heatwaves and convective phenomena. Long-term meteorological records (1991–2020) provided by the LAMMA Consortium (*Climatologia di Firenze 1991–2020*, n.d.) indicate an annual mean temperature value of 15.5 °C, which

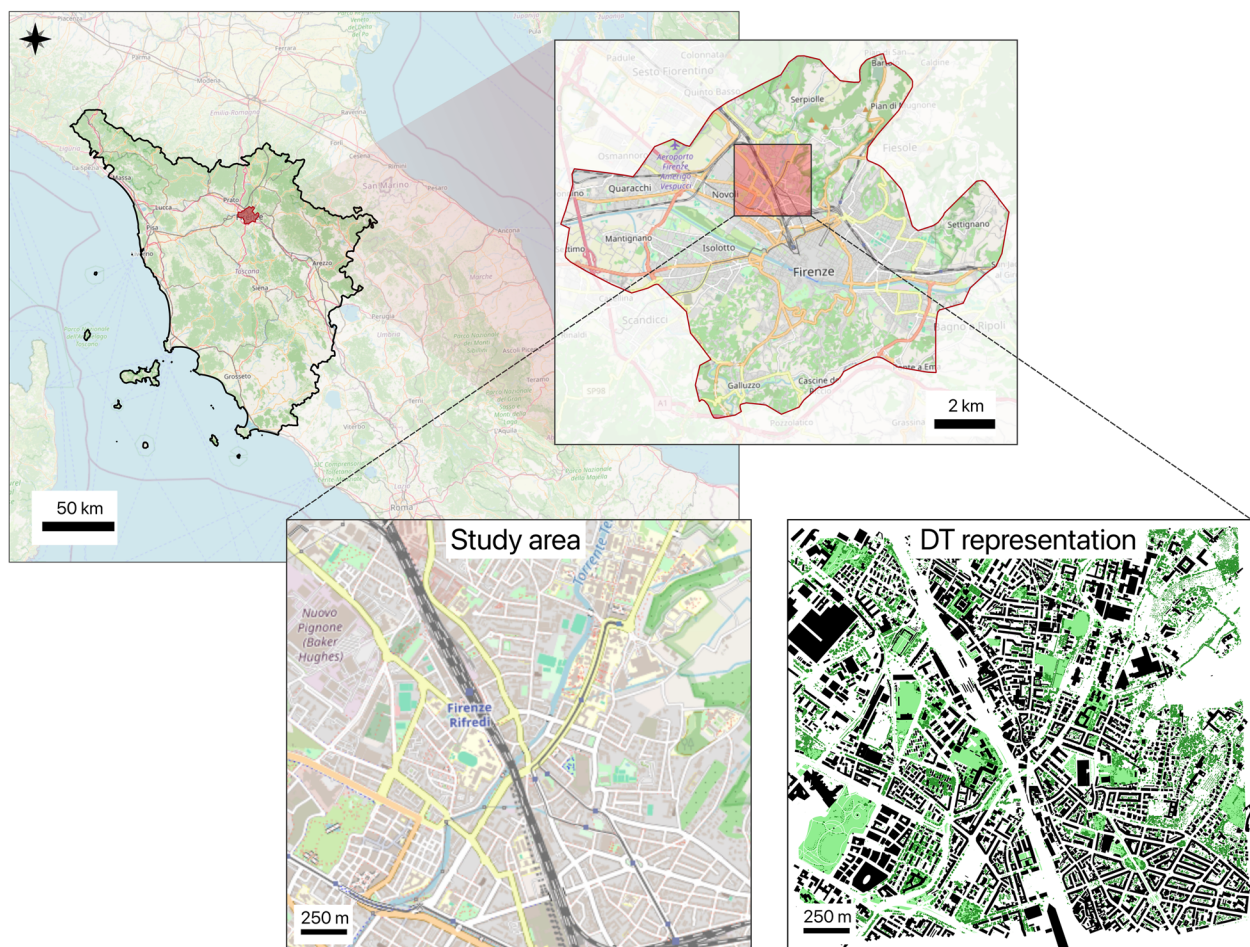


Fig. 1 View of Florence and its location in Tuscany Region, with a zoom on the study area (bottom right) and reconstruction of digitalized version for LUCIDit simulation with a 1 × 1 m resolution (bottom left). Black colour shows the built-up areas, light-green areas, greenery on the ground, and in dark green, the trees canopy. White areas are considered as asphalt

risers to a seasonal average of 24.5 °C during the summer months. Historical temperature extremes for the city range from a minimum of −10.2 °C to a maximum of 41.3 °C. Relative humidity typically fluctuates around 70%, peaking at 80% in winter and reaching its minimum average (approximately 65%) in July. Prevailing winter winds are predominantly south-westerly (SW), with average velocities ranging between 10 and 12 km/h.

This combination of intense summer solar radiation conditions and dense historical-to-modern urban morphology strictly connected to important radiative heat exchanges, makes Florence a critical pilot city for assessing people thermal stress and identifying the most effective targeted mitigation strategies.

2.2 High-resolution spatial and temporal mapping of MRT

To generate the high-resolution spatial data required for the ML framework, the domain is simulated using LUCIDiT, a physically based modelling architecture designed for rapid MRT assessment in complex urban environments (Baia et al., 2026). Rather than relying on computationally expensive CFD, the model operates on a simplified balance of mutual radiative heat exchanges.

The spatial discretization of the 5.5 km² study area in Florence was performed by integrating integrated public data provided by OpenStreetMaps (OSM) (*OpenStreetMap*, n.d.) with Digital Surface Model (DSM) to extract vegetation heights. Regarding ground surfaces, the domain adopts a binary classification based on the available data: vegetated areas (grass) and non-vegetated areas (which default to asphalt characteristics in this work). Consequently, water bodies were not explicitly digitized as a distinct class and were automatically categorized under the asphalt-like surface class during the initial preprocessing phase. While this simplifies water-specific latent heat exchanges, its macro-scale impact on the overall radiative clustering remains negligible due to the small spatial footprint of water bodies within the studied domain.

The LUCIDiT implementation used in this study results in two basic improvements and novelties: the thermal inertia assessment in a simplified way, and the modelling of greenery as a natural system calculating the evapotranspiration phenomena by the Penman–Monteith method (Grimmond & Oke, 1991; Monteith, 1965). Specifically, the complex transient heat transfer within different urban materials (e.g., asphalt, concrete) is parameterized using a recursive time-series filtering technique. By applying an Exponential Moving Average (EMA) (Nicol & Humphreys, 2010) to the radiative forcing, the model successfully mimics the delayed thermal response and nocturnal heat release of the built environment while reducing computational times by orders of magnitude.

Furthermore, the mitigation of radiative heat exchange, provided by greenery presence, is not limited to geometric shadow casting. Unlike purely geometric radiative models, LUCIDiT solves the comprehensive surface energy balance for each grid cell, explicitly accounting for latent heat fluxes and evapotranspiration processes of urban vegetation. In this simulation, urban trees are modelled as a homogeneous macro-species sharing identical structural and thermophysical parameters (including leaf transmissivity, albedo, and stomatal resistance). The evapotranspiration rates are dynamically calculated using the Penman–Monteith equation based on atmospheric forcing. However, since no hydrological soil model or explicit plant hydraulics are coupled, these rates do not depend on variable soil moisture, and photosynthetic activity is not considered. This simplified energy balance is significant and sufficient to capture the essential radiative shading and longwave canopy emission cooling without over-parameterizing the system.

The physical model used for data generation has already undergone rigorous empirical and site-specific validation against field measurements and established microclimates software in a previously peer-reviewed published paper (Baia et al., 2026), yielding highly robust results ($R^2=3.6$ °C, $MBE=-0.9$ °C comparing LUCIDiT results with UMEP model; $R^2=-0.3$ °C, $RMSE=5.6$ °C comparing the results with drone measurements).

MRT is calculated at a standard pedestrian level of 1.5 m above the ground, integrating the 3D shortwave (direct, diffuse, and reflected) and longwave radiation fluxes. An accurate description of the term considered and the model validation on real data and literature models is reported in (Baia et al., 2026).

The meteorological data used to drive the digital twin are extracted from the Typical Weather Year (TWY) representative of the 2023 local climate baseline (*Photovoltaic Geographical Information System (PVGIS)—Joint Research Centre*, 2024). To evaluate the urban morphologies under severe thermal stress, the simulation is configured for a 5-day summer period, ranging from August 10th to August 14th, solved with a temporal resolution of 1 h. This specific temporal window is selected to encompass the hottest day of the year, determined by the absolute maximum dry-bulb air temperature recorded in the dataset.

The simulation results are spatially continuous, hourly MRT maps with 1 × 1 m resolution (see Fig. 1). To ensure the physical reliability of the data fed into the clustering algorithm, a data pre-processing protocol is applied. To initialize the recursive EMA filters and stabilize the thermal inertia calculations, the first 24 h of the simulation (corresponding to August 10th) are excluded from the analysis. Consequently, the final multidimensional dataset used to train and validate the clustering algorithms

consists of a spatial grid where every single pixel is characterized by a continuous temporal sequence of 96 hourly values of MRT.

3 The method

This section details the defined methodological approach to translate continuous, multidimensional time series of MRT into discrete and easily usable urban "thermal morphologies".

To enhance the clarity of the experimental workflow, a comprehensive flowchart showing the overall research design and algorithm selection is provided in Fig. 2. Starting from the raw MRT time series, two key microclimatic indicators are extracted. They are subsequently used to define the physics-based heuristic baseline and to validate the thermophysical proprieties of the data-driven clustering. Using standard clustering evaluation metrics, the statistical performance is compared among the heuristic baseline, the raw time-series pipelines and the featured-engineered pipeline to select the optimal algorithm. The downstream validation and spatial translation phase are then executed by using this selected algorithm.

The remainder of this section is structured as follow: the heuristic method defined to establish a physical and thermodynamic robust baseline is detailed in Sect. 3.1. The selected unsupervised ML algorithms are introduced in Sect. 3.2, while Sect. 3.3 configure them into distinct experimental pipelines to compare their performances on row continuous data versus engineered features. In Sect. 3.4, the hyperparameter tuning regarding the number of clustering (k) is evaluated. Lastly, Sect. 3.5 introduces the novel 2D Thermal Phase Space and the microclimatic indicators used to physically validate the clustering results.

3.1 Taxonomy framework based on classical physics and thermodynamics

A deterministic rule-based framework is developed to provide a physical and thermodynamic strong basis and translate the raw time series of MRT into real microclimatic information content. This heuristic approach relies on a decision tree constructed from two fundamental thermodynamic indicators: the Global Thermal Range ($\Delta T_{max-min}$) and the Thermal Asymmetry Index (ΔT_{PM-AM}). The global thermal range captures the total amplitude of the daily thermal wave, and the thermal asymmetry index quantifies the directional phase shift caused by specific solar exposures.

To establish these parameters, the morning and afternoon MRT average values are computed over specific time windows under clear-sky conditions, defined as AM (09:00–12:00) and PM (14:00–17:00) local time. Consequently, the Thermal Asymmetry Index is formally defined as the difference between the PM and AM averages (ΔT_{PM-AM}).

The numerical peak hour (h_{peak}) is strictly constrained between 08:00 and 19:00, due to all the numerical artifacts elimination and the evaluation of nocturnal thermal re-radiation towards the Sky.

Indeed, the Global Thermal Range ($\Delta T_{max-min}$) is computed over the full 24-h cycle to accurately capture the nighttime cooling phase, which is heavily governed by the local SVF.

To partition the urban fabric into discrete thermal morphologies, explicit thresholds are defined, based on the statistical distribution of the dataset. Specifically, the threshold for continuously shaded areas is dynamically set at the 25th percentile (P_{25}) of the global thermal range distribution. This threshold assumes that the 25% of the thermal profiles correspond to locations with high physical shielding, such as deep urban canyons or dense

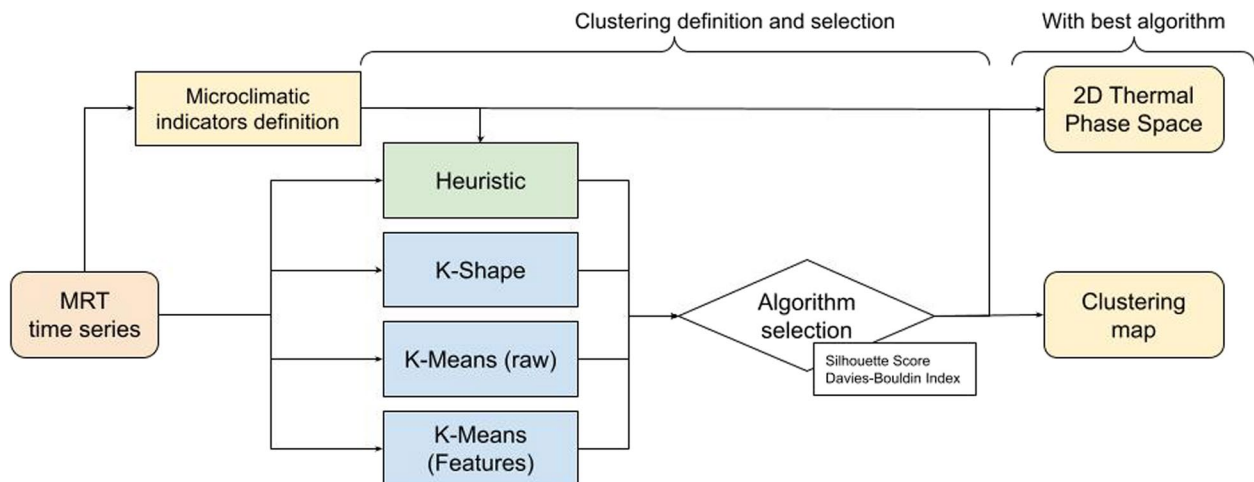


Fig. 2 Flowchart of the proposed method, from input MRT time series to the dual-layer actionable planning map

vegetation. For sun-exposed areas, directional exposure is classified using a predefined asymmetry threshold of $\pm 2^{\circ}\text{C}$. The heuristic rules classify pixels as shadow or vegetation if their thermal range is below the 25th percentile. If the range exceeds this threshold, pixels are categorized as East exposure when the thermal asymmetry index is less than or equal to -2°C , and as West exposure when the index is greater than or equal to 2°C . Pixels exceeding the thermal range threshold but exhibiting an asymmetry index between -2°C and 2°C are classified as open (or symmetric) surfaces. This deterministic approach uses rigid boundaries that are difficult to adapt to nonlinear transitions and even impulsive changes typical of the microclimate of complex urban morphologies. However, it is consistent with real physical phenomena.

3.2 Unsupervised machine learning framework

An unsupervised ML is employed to autonomously identify thermal morphologies based on the intrinsic data distribution without relying on predefined physical labels. This allows the overcoming of limitations of deterministic thresholds. Two distinct clustering algorithms are selected for comparison: the K-Means algorithm and the K-Shape algorithm.

The K-Means algorithm is a classic partition-based clustering method designed to divide the dataset into a predefined number of non-overlapping subsets. The algorithmic procedure begins by initializing k centroids within the multidimensional space. During the assignment phase, each continuous time series is allocated to the cluster with the nearest centroid. This proximity is mathematically defined by minimizing the within-cluster variance using the Euclidean distance. Subsequently, the update phase recalculates the position of each centroid by computing the mean of all the time series assigned to that specific cluster.

The iterative process of assignment and update continues until convergence is reached, meaning the cluster assignments no longer change. Because the Euclidean distance strictly evaluates the amplitude difference between data points at corresponding, synchronized time steps without shifting the signal, K-Means inherently preserves the temporal phase of the thermal peaks. For the microclimate analysis, this strict temporal alignment makes the algorithm well-suited for capturing directional solar dynamics distinguishing a morning radiative peak from an afternoon one.

The K-Shape algorithm is a specialized time-series clustering method explicitly engineered to preserve the geometric profiles of data sequences, regardless of potential phase shifts. Unlike K-Means, K-Shape relies on a shape-based distance metric derived from cross-correlation. During the assignment phase, the algorithm evaluates the similarity between a time series and a cluster

centroid by shifting the sequence backward and forward along the time axis to find the optimal alignment that maximizes their cross-correlation. Once the optimal shift is determined, the series is assigned to the cluster sharing the most similar morphological pattern. The centroids are then updated using an optimization framework that extracts the most representative underlying shape from the aligned sequences.

While this shift-invariant property makes K-Shape highly effective in domains where the shape of a signal is more important than its timing, it introduces a physical limitation in urban climatology. By artificially aligning the time series to maximize geometric similarity, this phase-shifting process neglects the temporal heat load of solar exposure, which is strictly governed by the absolute time of day rather than merely the shape of the heating curve.

3.3 Data preprocessing and scaling strategies

Clustering algorithms are tested under different input conditions to identify the optimal classification strategy. Therefore, three distinct ML pipelines are identified and compared with the reference heuristic method. The first pipeline applies the K-Shape algorithm directly to the continuous multi-day MRT profiles to assess the viability of shape-based grouping in urban microclimate. The raw MRT time series are normalized using a Mean-Variance Scaler (Z-score normalization). This standardization transforms each individual time series to have a zero mean and unit variance. This step is mathematically required for K-Shape, as its cross-correlation-based distance metric is engineered to evaluate pure geometric structural similarity (e.g. the “shape” of the thermal wave), independent of absolute temperature amplitudes and baseline shifts. Due to the extreme computational complexity of the Shape-Based Distance metric, the K-Shape algorithm is fitted on a uniformly distributed random subset of 100k time series. The resulting centroids are then used to assign the remaining pixels of the full district domain. This sample size, representing approximately 2.5% of the total dataset, is statistically and thermodynamically sufficient to capture the full heterogeneity of the domain. Indeed, because the urban built-up area is characterized by recurring geometric and exposure patterns, a uniform sample of this magnitude, randomly extracted, ensures that all the different time series configuration are represented. This allows to fully stabilize the cluster centroids without heavy computational effort.

The second pipeline applies the K-Means algorithm to a pre-processed dataset containing exactly nine thermal features. This pipeline is designed to ascertain whether human-guided dimensionality reduction and physics-based feature engineering ultimately improve the overall

clustering accuracy. To systematically extract these indicators, the multidimensional time series data are first aggregated to compute a representative mean daily thermal profile for each spatial pixel. The extracted scalar features are mathematically designed to capture specific physical dimensions of the local microclimate. Thermal amplitude is parameterized through the absolute maximum temperature, the absolute minimum temperature, and the resulting diurnal thermal range. To accurately characterize the temporal phase of the radiative forcing, a temporal centroid (t_c) is calculated for each pixel. This centroid is defined as the weighted average of the daily hours, i.e. weighted by the difference between the existing Mean Radiant Temperature ($\overline{MRT}_h$) and their mean daily minimum value ($\overline{MRT}_{min}$):

$$t_c = \frac{\sum_{h=0}^{23} (\overline{MRT}_h - \overline{MRT}_{min}) \cdot h}{\sum_{h=0}^{23} (\overline{MRT}_h - \overline{MRT}_{min})}. \quad (1)$$

The peak delay is computed to measure the exact temporal shift between the local pixel's maximum and the global spatial average peak hour. The geometric asymmetry of the thermal wave is captured by computing the statistical skewness of the temporal distribution, which is then averaged across all simulated days.

To quantify the localized cumulative heat stress, a thermal load integral is calculated by summing all the temperatures that exceed a dynamic threshold, established at the 75th percentile of the typical daily profile. The transient dynamics of energy accumulation and dissipation are evaluated through specific temporal gradients: the morning heating gradient, calculated as the linear temperature variation between 06:00 and 10:00, and the evening cooling gradient, calculated between 16:00 and 20:00.

The third pipeline applies the K-Means algorithm directly to the raw, unfiltered continuous MRT time series.

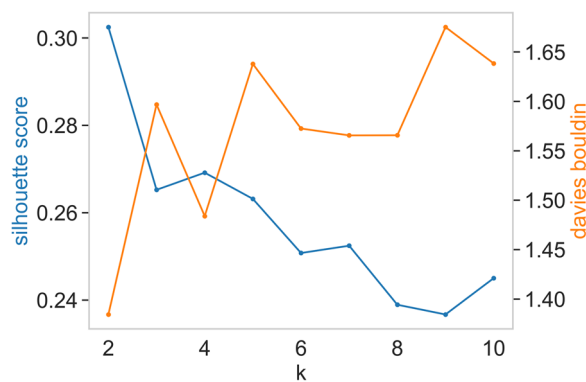


Fig. 3 Hyper-parameter tuning analysis evaluating cluster numbers (k) from 2 to 10

In particular, the standardizing features is a common practice in conventional ML to prevent variables with larger numerical ranges from dominating the objective function, but for the present study and in this specific context it cannot be used. In the continuous MRT dataset, all the temporal features (i.e. discrete time steps) share the exact same physical unit ($^{\circ}\text{C}$) and scale.

For thermodynamics, the standardization of MRT series would cancel the physical identity of the pixels. For instance, forcing a zero mean and unit variance would produce the same thermal profile of a persistently shaded urban canyon (characterized by a low baseline and minimal amplitude) as an urban square completely exposed to solar radiation (characterized by a high baseline and extreme daytime peaks). Using the Euclidean distance on the raw, unscaled MRT data, the K-Means algorithm is mathematically forced to implicitly evaluate both the temporal shape (phase shifts due to orientation) and the absolute thermal load (the raw amplitude of the thermal radiative forcing). The strategic choice not to use standardization allows, by means of the algorithm, to successfully separate thermo-physical and microclimatic archetypes and not purely geometric patterns.

3.4 Internal validation and hyper-parameter tuning

In this section the internal validation method and hyper-parameter tuning or optimization, for the used learning algorithm control with ML, is described and discussed.

An internal validation loop is executed evaluating k values from 2 to 10, to determine the optimal number of thermal archetypes (k) for the study urban domain. The performance of the K-Means algorithm on the raw continuous MRT series is assessed using two standard clustering metrics: the Silhouette Score (to measure intra-cluster cohesion and inter-cluster separation), and the Davies-Bouldin Index (to evaluate the ratio of within-cluster scatter to between-cluster separation).

As shown in Fig. 3, the mathematical optimization points towards $k=2$ as the absolute optimum across all the metrics (e.g., maximizing the Silhouette Score and minimizing the Davies-Bouldin Index). From the thermodynamics point of view, this binary split ($k=2$) flawlessly captures the primary physical and thermo-physical variance of the urban environment: the fundamental dichotomy between continuously irradiated surfaces (open spaces) and physically shielded environments (shadow areas/canyons).

However, while mathematically robust, a binary classification would be ineffective and insufficient for a sustainable resilience urban planning. Indeed, it completely masks the well-known secondary critical variance governed by the street network orientation in respect to the solar path from sunrise to sunset which manifests itself in

the time-lag difference between morning and afternoon thermal peaks.

For this reason, a local optimum of $k=4$ is selected. As demonstrated in the subsequent 2D Phase Space analysis (Sections. 3.5 and 4.4), $k=4$ provides the optimal choice and equilibrium between statistical cluster validity and domain-specific physical meaning, successfully isolating the different thermodynamic and thermo-physical behaviours of East and West exposures. To ensure a rigorous methodological comparison, this physically validated parameter is consistently applied across all the unsupervised learning algorithms tested for this study.

3.5 Microclimatic indicators for validation

To rigorously validate the physical consistency of the unsupervised clustering results, a diagnostic 2D Thermal Phase Space is established. This space is constructed by utilizing the two thermodynamic indicators previously defined in Sect. 3.1—the Thermal Asymmetry Index (ΔT_{PM-AM}) and the Global Thermal Range ($\Delta T_{max-min}$)—as orthogonal axes. The conceptual framework of this phase space is designed to capture the fundamental feature of the thermal signature of any urban pixel, regardless of its spatial coordinates. The y-axis ($\Delta T_{max-min}$) serves as a proxy for the thermal intensity and thermal radiative forcing, separating shielded environments from open, thermal high-amplitude surfaces. The x-axis (ΔT_{PM-AM}) represents the temporal phase shift of the thermal wave, effectively identifying the directional orientation (East vs West) based on the timing of the maximum solar heat load.

By projecting the multi-dimensional results of the K-Means and K-Shape algorithms into this bi-dimensional physical domain, it is possible to assess whether the unsupervised clusters align with the expected thermodynamic quadrants. This mapping acts as a physical ground truth that allows for a quantitative and qualitative evaluation of the ML framework, ensuring that the identified thermal urban morphologies are not numerical artifacts but constitute distinct urban microclimates, thermo-physically consistent and representative of the real conditions.

4 Results and discussion

4.1 Computational efficiency and algorithm performance

The operational viability of any urban microclimate assessment tool heavily depends on its computational scalability, especially when transitioning from neighbourhood-scale to continuous district-scale, or whole cities, applications. In this study, the classification framework processed an extensive spatial domain of 5.5 km², translating to over 4 million of MRT time series.

Table 1 summarizes the total execution times required by each approach, inclusive of all necessary data pre-processing and feature engineering phases, executed on an Apple Silicon M3 Pro processor and 18 GB of unified memory.

As expected, the deterministic heuristic approach exhibits the lowest computational cost (~5.4 s), given its reliance on rapid vectorized thresholding operations. Among the ML algorithms, K-Means applied directly to raw time series demonstrates remarkable efficiency (~24.1 s), dramatically outperforming both K-Means with engineered features (~73.8 s) and the K-Shape algorithm (~15 min). The computational overhead of K-Shape is primarily driven by the mathematical complexity of its cross-correlation calculations (Shape-Based Distance) across massive datasets.

To make the execution of K-Shape computationally tractable within a reasonable acceptable timeframe, the model is fitted on a randomly extracted subset of 100k time series, before extending the centroid-based classification to the rest of the dataset.

The fact that K-Shape requires several minutes despite this severe spatial down-sampling during the training phase highlights the unsuitability of shape-based distances for continuous district-scale applications. Therefore, the raw K-Means approach turns out as the best computational ML solution, completing the multi-day spatial classification of an entire district in under half a minute. However, raw computational speed must be strictly balanced against physical accuracy. While the baseline heuristic method is five times faster, its rigid theoretical structure heavily compromises the reliability of the classification, as explained in detail in the following section.

4.2 Physical limitations of the heuristic baseline

The rule-based heuristic approach generally provides a near-instantaneous execution, but its deterministic nature inherently struggles with the physical continuity and complexity of the real urban fabric. The complex geometry of the 5.5 km² district (characterized by varying canyon aspect ratios, staggered facades, and irregular greenery) produces continuous, non-linear microclimatic transitions that cannot be adequately captured by static global thresholds.

Table 1 Execution times for each algorithm

Classification method	Input data strategy	Total execution time
Heuristic	Precomputed indicator	5.4 s
K-Means (Raw)	Continuous time series	24.1 s
K-Means (Features)	Engineered scaled features	73.8 s
K-shape	Standardized time series*	15 min

*100 k subset fitting + full assignment

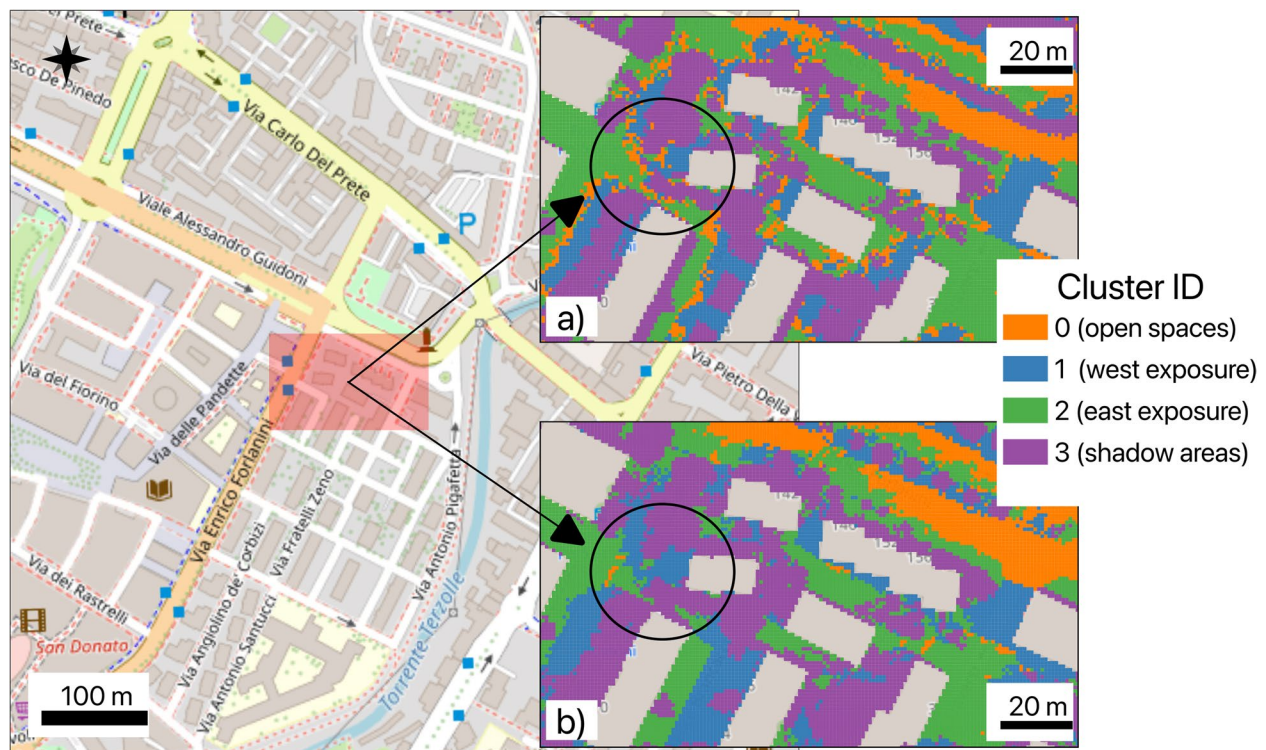


Fig. 4 Detail of the spatial clustering obtained in a sub-area of the target domain obtained with the (a) heuristic baseline versus (b) K-Means

Table 2 Silhouette Score and Davies-Bouldin Index for the proposed approach

Algorithm	Silhouette Score	Davies-Bouldin Index
Heuristic	0.06	2.45
K-Means (Raw)	0.27	1.47
K-Means (Features)	0.18	3.05
K-shape	0.15	2.34

As explained in Sect. 3.1, the baseline relies on rigid cut-offs, such as the exact 25th percentile for shadow areas or the strict $\pm 2^\circ\text{C}$ limit for directional exposure. This discrete partitioning produces significant classification artifacts in transitional boundary zones. For instance, a physical space experiencing a thermal asymmetry of 1.95°C is artificially forced into the symmetric “open space” category, while an adjacent, practically identical pixel at 2.05°C is abruptly classified as “west exposure”.

Visualizing the spatial output of the heuristic method (Fig. 4) reveals these critical structural limitations. The resulting categorical map is characterized by fragmented boundaries and unnatural “salt-and-pepper” artifacts, particularly around the shadow edges of urban canyons and tree canopies. These rigid boundaries demonstrate that a pure deterministic tool despite its unmatched computational speed, fails to capture the thermodynamics synergistic effects and the smooth thermal gradients, the knowledge of which is fundamental for reliable and

sustainable resilience urban planning. This fundamental limitation necessitates the shift towards the dynamic, variance-driven learning provided by the K-Means algorithm, which progressively adapts to the data.

4.3 Algorithm selection and clustering validation

The first objective of the analysis is to determine the most robust and mathematically consistent method for classifying urban thermal dynamics. Table 2 summarizes the internal validation metrics—specifically the Silhouette Score and the Davies-Bouldin Index—evaluated across the baseline heuristic approach and the three unsupervised ML pipelines.

To properly contextualize the performance of the tested algorithms, it is essential to define the theoretical bounds and practical implications of these metrics.

The Silhouette Score ranges from -1 to 1 and quantifies the balance between intra-cluster cohesion and inter-cluster separation. According to literature (Rousseeuw, 1987), values above 0.5 indicate well-separated clusters, whereas values between 0.25 and 0.5 may still reveal meaningful underlying structure, particularly in datasets characterized by gradual transitions rather than discrete categories. Urban microclimatic fields belong to this latter class, as thermal conditions change continuously across space, due to progressive variations in shading, SVF, and radiative heat exchanges. Therefore, Silhouette Score values between 0.25 and 0.5 are expected even

when physically interpretable thermal regimes are correctly identified.

The Davies-Bouldin Index evaluates the ratio between, within-cluster dispersion and between-cluster separation, with lower values indicating superior clustering performance (Davies & Bouldin, 1979). Since no universal acceptance threshold exists for this metric, this index is primarily employed here as a non-absolute measure for comparing alternative clustering strategies applied to the same dataset.

The quantitative validation identifies the K-Means algorithm applied directly to the raw MRT time series as the optimal classification strategy, achieving the highest Silhouette Score and the lowest Davies-Bouldin Index among all the tested methods. Although the absolute Silhouette value remains moderate (0.27), it is the highest among all the tested approaches. Due to the continuous nature of urban microclimatic fields, where thermal regimes transition gradually across space rather than forming sharply separated categories, these Silhouette values can be expected. This happens even when physically meaningful thermal structures are present. At the same time, the Davies-Bouldin Index reaches its minimum value (1.47), confirming the best compromise between compactness and separation among the evaluated methods.

In stark contrast, the deterministic heuristic approach exhibits the poorest mathematical performance, yielding a Silhouette Score near zero (0.06) and a comparatively high Davies-Bouldin Index (2.45). These metrics mathematically confirm that rigid, user-defined thresholds generate highly overlapped clusters with weak internal consistency. The alternative ML pipelines perform intermediately but remain sub-optimal. K-Means applied to the engineered features records a Silhouette Score of 0.18 but suffers from the highest Davies-Bouldin Index (3.05), indicating severe dispersion and poor separation within the reduced feature space.

Similarly, the K-Shape algorithm applied to the raw series yields inadequate metrics (Silhouette Score of 0.15; Davies-Bouldin Index of 2.34), suggesting that a purely shape-based distance metric struggles to cleanly partition the temporal variance of the dataset.

Beyond these pure statistical metrics, the alternative methods exhibit critical physical limitations when forced to extract four discrete microclimatic classes. While the heuristic approach successfully anchors the classification to classical physical thresholds, it produces rigid, fragmented spatial boundaries that inherently fail to capture the continuous thermodynamic transitions characterizing complex urban canyons. The K-Shape algorithm, applied to the raw series, suffers critically from its shift-invariant nature. By artificially aligning the delayed thermal peak of West-facing surfaces with the early peak

of East-facing surfaces to maximize cross-correlation, K-Shape completely loses the fundamental temporal signature of solar exposure. Consequently, it merges geometrically opposed and thermodynamically distinct areas into physically ambiguous clusters. Furthermore, reducing the 96-h continuous waveform into nine discrete scalar features for the K-Means algorithm results in a significant loss of the holistic temporal dynamics.

This manual dimensionality reduction not only leads to the aforementioned mathematical instability, but also generates a noisier, less coherent spatial distribution across the urban domain.

Due to its higher mathematical performance and its intrinsic algorithmic ability to preserve the synchronization of temporal peaks (i.e. a feature essential for accurately capturing directional solar radiation), the K-Means algorithm, directly applied to the raw time series, is selected as the core model for the microclimatic analysis. Consequently, the subsequent physical validation and spatial translation phases will focus exclusively on the four thermal archetypes extracted by this optimal framework.

4.4 Microclimatic signatures and thermal phase space

Having confirmed the K-Means algorithm applied to raw time series as the optimal classifier, the physical meaning of the extracted clusters is thoroughly investigated. Crucially, without any prior rule-based instruction or data simplification, the unsupervised algorithm autonomously identifies four distinct thermal archetypes that mirror the well-known urban thermodynamics and its thermophysical behaviour: shadow areas, east-facing surfaces, west-facing surfaces, and completely open surfaces. To quantitatively validate the thermodynamic consistency of these data-driven clusters, the spatial pixels are projected into a novel two-dimensional Thermal Phase Space (Fig. 5). This phase space is mathematically defined by the Global Thermal Range ($\Delta T_{max-min}$) on the y -axis, representing the absolute amplitude of the daily thermal wave, and Thermal Asymmetry Index (ΔT_{PM-AM}) on the x -axis, capturing the directional temporal shift of the radiative forcing.

As illustrated in Fig. 5, the K-Means algorithm naturally segregates the urban fabric into distinct areas of the phase space, effectively reconstructing the physics of solar exposure purely from the intrinsic temporal variance of the data. Specifically, Cluster 0 (open surfaces) is characterized by the highest global thermal range (mean $\Delta T_{max-min} \sim 52.8^\circ\text{C}$) while exhibiting negligible thermal asymmetry ($\Delta T_{PM-AM} \sim 0^\circ\text{C}$). This thermodynamic signature is fundamentally representative of horizontal surfaces, such as wide-open squares with high SVE, which receive symmetric solar irradiation throughout the diurnal cycle. Conversely, the algorithm

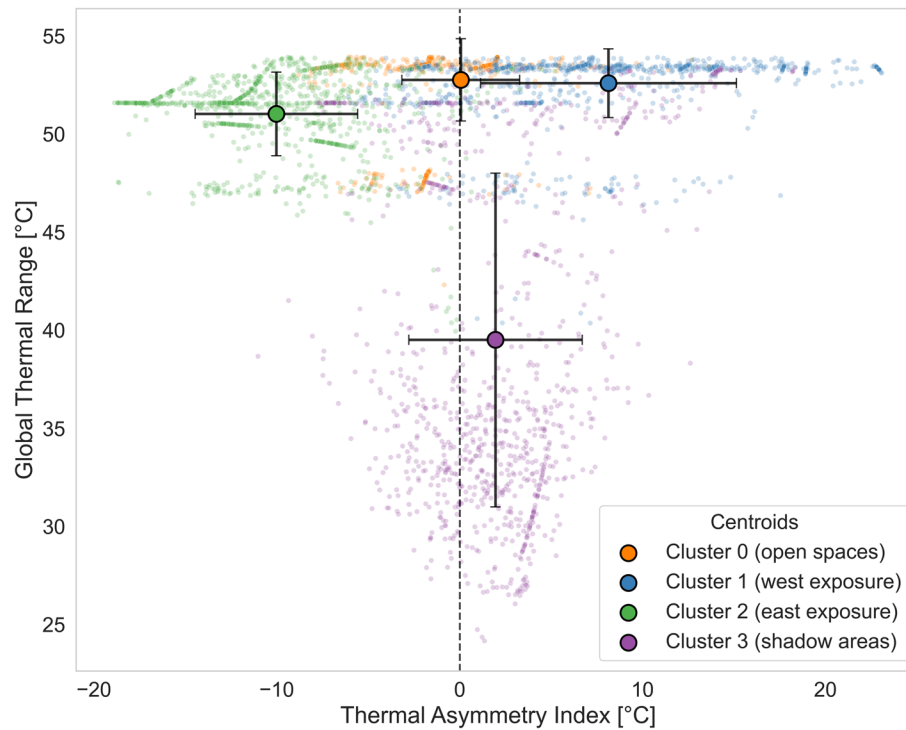


Fig. 5 Thermal phase space. Larger dot with error bars indicates the mean value and the standard deviation of the two microclimatic indicators for each cluster. Smaller dots represent a random sample of 1000 elements for each cluster

Table 3 Microclimatic indicators for the four clusters extracted with K-mean application on raw data

Cluster	Area [%]	$\Delta T_{max-min}$	ΔT_{PM-AM}	t_{peak}
0: open surface	31.68	52.8	0.0	14:22
1: W/SW exposure	21.32	52.6	8.1	14:26
2: E/NE exposure	20.29	51.0	-10.2	12:32
3: shadow area	26.72	39.5	2.0	12:37

clearly separates the directionally polarized environments along the horizontal axis. Cluster 1 (west exposure) exhibits both a high thermal range and a strong positive asymmetry ($\Delta T_{PM-AM} \sim 8.1^\circ\text{C}$), perfectly capturing the delayed radiative peak of the afternoon sun, a heat stress condition that is typically exacerbated by the concurrent accumulation of high background air temperatures. On the opposite spectrum, Cluster 2 (east exposure) displays a comparably high thermal range but a strong negative asymmetry ($\Delta T_{PM-AM} \sim -10.2^\circ\text{C}$), physically corresponding to vertical surfaces that rapidly heat up during the early morning hours but are effectively shielded by the surrounding urban geometry during the afternoon peak. Finally, Cluster 3 (shadow and vegetation) collapses into the lower-centre portion of the phase space. This specific archetype successfully isolates pixels with significantly dampened thermal ranges (mean $\Delta T_{max-min} \sim 39.5^\circ\text{C}$) and minimal asymmetry, a subdued thermal behavior typical of highly shielded environments, such as deep urban canyons or dense tree

canopies, where direct shortwave radiation is persistently restricted. The exact statistical distributions and mean microclimatic properties of these validated archetypes are comprehensively summarized in Table 3.

4.5 Physical and morphological characterization of the classes

To provide a comprehensive understanding of the four K-Means clusters, their temporal, thermodynamic, and morphological signatures are systematically analysed. Figure 6 illustrates the mean diurnal MRT profiles for each cluster for the hottest and most characteristic day of the dataset (13 August 2023). The temporal shift in peak temperatures perfectly differentiates the directional classes: Cluster 2 (east exposure) exhibits a steep morning heating gradient terminating in an early peak, whereas Cluster 1 (west exposure) demonstrates a significantly delayed thermal peak driven by the progressive accumulation of afternoon solar irradiance. This directional polarization is statistically confirmed by the MRT asymmetry distributions (Fig. 6 b), which clearly isolate the strong positive asymmetry of the West exposition and the strong negative asymmetry of the East exposition, highlighting the strict dependence of these thermal waves on the solar path.

The purely data-driven thermal classification is explicitly cross-validated against the physical urban morphology using localized spatial View Factors (Fig. 7). Within

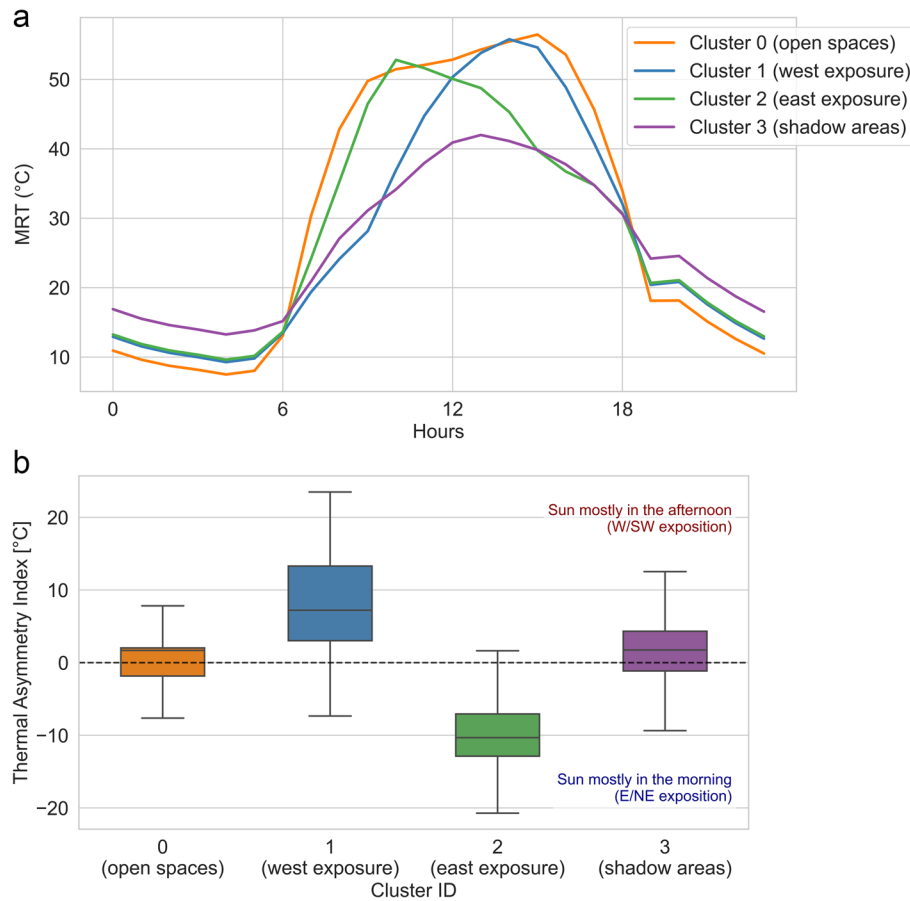


Fig. 6 **a** Mean MRT time series of the hottest day and **b** Thermal Asymmetry Index distribution for the four classes

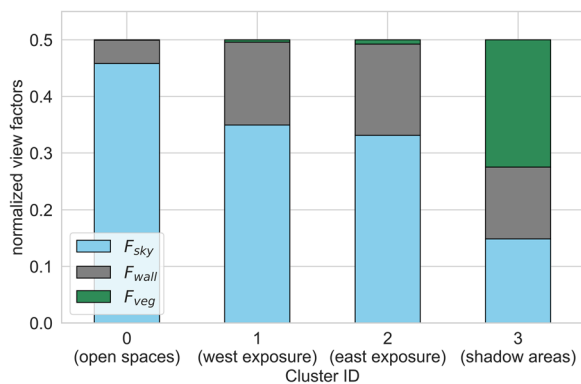


Fig. 7 Normalized view factor (F_{sky} , F_{wall} , F_{veg}) distribution for each cluster

the established LUCIDiT modelling framework, the radiative environment of each computational pixel is geometrically governed by specific view factors (F_i), which quantify the fraction of the surrounding hemisphere occupied by different morphological elements.

Specifically, F_{sky} represents the unhindered view of the celestial vault, governing both direct shortwave irradiance and nocturnal longwave cooling. F_{wall} quantifies the solid angle subtended by vertical building facades, acting

as thermal mass and secondary radiative emitters, while F_{veg} accounts for the physical shielding provided by tree canopies and green infrastructure, as thoroughly detailed in (Baia et al., 2026).

This morphological cross-validation reveals a profound physical consistency between the algorithm's thermodynamic clusters and the actual three-dimensional city structure. As illustrated in Fig. 7, Cluster 0 (open spaces) is strongly dominated by a high SVF (F_{sky}), physically corroborating its unhindered exposure to solar radiation and its consequent maximum daily thermal range. In contrast, Clusters 1 and 2 (west and east exposures, respectively) are characterized by a predominant fraction of the Wall View Factor (F_{wall}). This morphological signature confirms that these archetypes correspond to the vertical facades and intra-canyon street networks, where the fundamental distinction between the two classes is governed not by their geometric enclosure, but exclusively by their astronomical orientation relative to the sun. Finally, Cluster 3 (shadow areas) is characterized by the lowest F_{sky} and the highest combined fractions of wall (F_{wall}) and vegetation (F_{veg}) view factors. This severe physical shielding perfectly explains the heavily dampened MRT profile, and the lowest daily thermal range captured by the

clustering algorithm, confirming that dense urban canyons and tree-covered areas behave thermodynamically as a unified, highly protected archetype.

4.6 Spatial translation for climate resilience urban planning

The ultimate objective of the proposed ML framework is to translate massive, multidimensional numerical MRT series into an intuitive, spatially explicit format for practical urban planning applications. To convert unsupervised temporal clustering result in practical tools for urban planners, the spatial visualization framework must retain information about both temporal trends and absolute thermal stress duration. A potential drawback of standard categorical clustering is the masking of absolute thermal magnitude. Pixels belonging to different temporal clusters might still experience comparable cumulative heat stress during peak hours, potentially misleading planners regarding intervention priorities. To overcome this limitation and provide an explicit metric for risk scale, the final classification map (Fig. 8) is updated by cross-referencing the K-means clusters with an absolute thermal threshold exceedance analysis. A critical MRT = 40 °C threshold is selected, and the cumulative number

of hours exceeding this value is computed for each pixel over the 96-h period. The selection of the critical threshold is grounded in established outdoor thermal biometeorology standards. Under typical summer heatwave conditions, MRT above 40 °C characterized unshaded areas with continuous Sun exposition. This threshold value acts as a physical driver that shifts comprehensive diagnostic indices, such as the PET, into the 'Strong Heat Stress' regime (PET > 35 °C), where radiative heat exchange imposes significant physiological strain on the human cardiovascular system (Amengual et al., 2014; Crandall & González-Alonso, 2010; Muthers et al., 2010).

The discretization of the temporal duration into Classes A–D follows an empirical, data-driven equal interval classification approach, based on the simulation boundaries. Although the total evaluation period spans 96 h, the diurnal solar cycle restricts this absolute maximum duration, during which any single pixel within the domain exceeds the 40 °C threshold, to 42 h. This value constitutes the number of hours during which the solar radiation is not zero. Consequently, the actual continuous distribution of heat stress duration ranges from 0 to 42 h. This empirical range is partitioned into four bins of approximately 10.5 h each, mapping onto successive

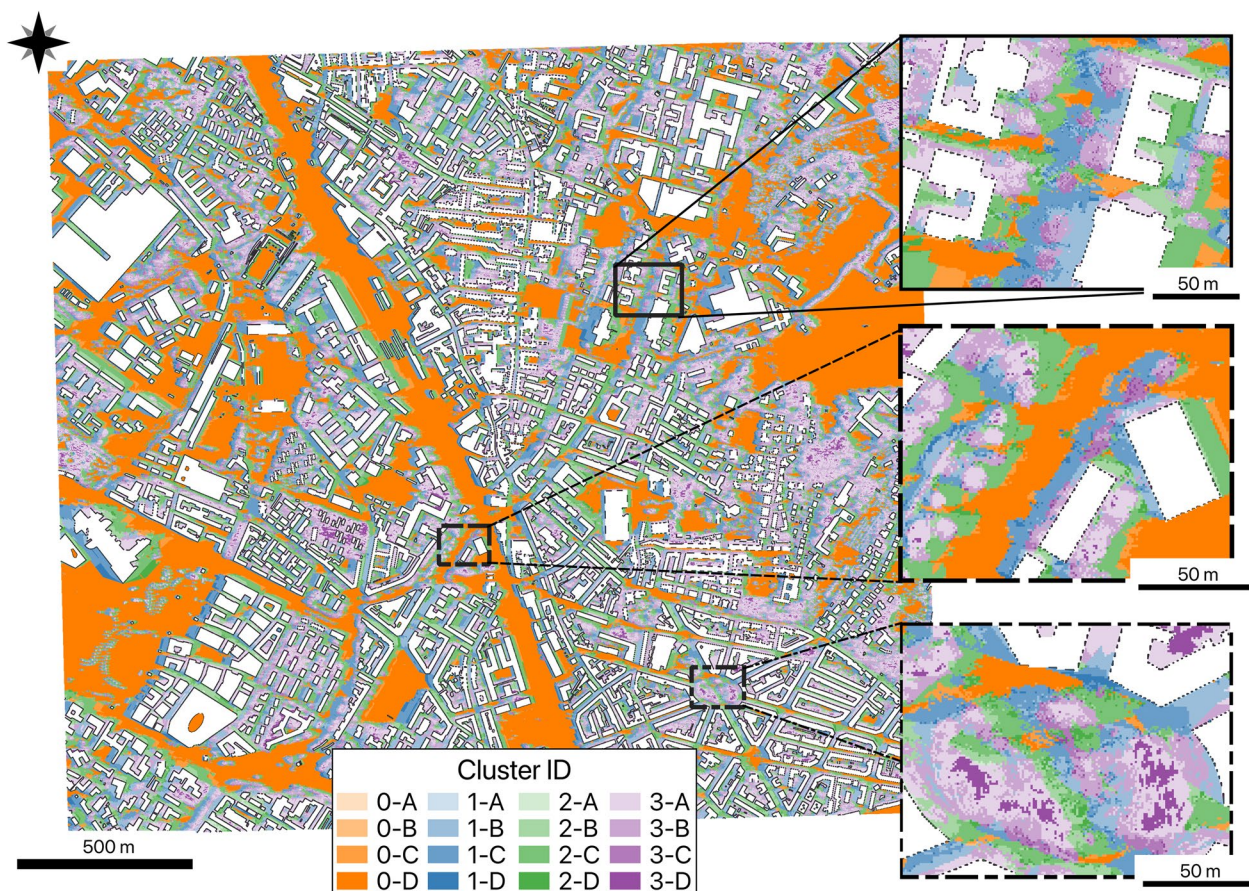


Fig. 8 High-resolution spatial distribution of the four extracted microclimatic cluster modulated by alpha-channel thermal stress opacity

Table 4 Percentage distribution of pixels within each microclimatic cluster across absolute thermal stress exceedance classes

Cluster	Class			
	A [%]	B [%]	C [%]	D [%]
0: open surface	0.00	0.07	9.16	90.77
1: W/SW exposure	0.22	42.35	48.28	9.15
2: E/NE exposure	2.17	31.87	54.30	11.65
3: shadow area	49.86	38.78	5.83	5.52

quartiles of stress exposure severity (0–25% for Class A, 25–50% for Class B, 50–75% for Class C, and 75–100% for Class D).

By mapping these classes onto the alpha channel (opacity) of the cluster colours, the updated visualization offers a dual-layer decision tool. The hue identifies the specific thermal morphology (suggesting how to mitigate, e.g., cool pavements vs. high-canopy trees), while the opacity explicitly flags high-priority hotspots (suggesting where to mitigate at first).

Remarkably, although the K-Means algorithm processes the time series purely based on their temporal variance (without any prior knowledge of spatial coordinates, building footprints, or geometric boundaries) the resulting categorical map reveals a highly coherent spatial distribution that strictly adheres to the physical urban morphology.

The spatial projection accurately maps Cluster 0 to continuously unshielded environments, flawlessly delineating wide-open squares, major intersections, and broad urban avenues. In stark contrast, Cluster 3 is meticulously localized within physically protected zones, such as dense tree canopies, narrow intra-block courtyards, and deep urban canyons where direct solar access is persistently restricted. Furthermore, the map vividly captures the directional polarization of the urban street network. Clusters 1 and 2 are systematically distributed along the respective sun-exposed facades and pedestrian sidewalks, perfectly tracing the astronomical orientation of the built environment.

The distribution of pixel percentages across these subclasses (Table 4) provides a quantitative validation of the dual-layer visualization framework. Remarkably, 90.77% of the pixels classified under Cluster 0 (open surfaces) falls into Class D, experiencing more than 32 h of critical thermal stress ($MRT > 40^{\circ}\text{C}$) over the 96-h simulation period. This confirms that unshielded urban nodes represent the most severe and persistent thermal condition.

Conversely, the directional archetypes (Cluster 1 and Cluster 2) display a highly intermediate distribution, peaking heavily within Class B and Class C. Specifically, Cluster 1 (W/SW exposure) exhibits 42.35% in Class B and 48.28% in Class C, while Cluster 2 (E/NE exposure) shows 31.87% in Class B and 54.30% in Class C.

Although their thermal heat stress durations are statistically comparable, their temporal profiles are entirely distinct (morning vs. afternoon exposure), reinforcing why duration metrics alone are insufficient without temporal clustering.

Finally, Cluster 3 (shadow areas) acts as a primary urban thermal buckler, with nearly half of its pixels (49.86%) confined to Class A (0–10 h of stress) and 38.78% in Class B, showcasing the crucial role of dense canopies and micro-scale structural shielding in minimizing heat exposure.

5 Discussion

5.1 Added value for actionable urban planning

The core innovation of the proposed machine learning framework lies in its ability to translate massive, multi-dimensional microclimatic data into a single, intuitive, and operationally actionable diagnostic map. Usually, urban climate planning heavily relies on static morphological proxies, such as the SVF or simple aerial photography, to identify potential urban heat hotspots. Recent literature reviews confirm a growing interest in leveraging artificial intelligence to connect urban morphology to microclimate mapping, but the used indicators include a structural limitation: orientation blindness.

An SVF map or an orthophoto can effectively separate wide-open spaces from deep architectural shadows, but it remains fundamentally invariant to solar trajectories and building facade orientation. Consequently, a general morphological analysis would inevitably combine East-facing and West-facing street canyons into a single category. To solve these distinct temporal regimes using traditional methods, urban planners would need to perform complex, computationally heavy geometric intersections between SVF and dynamic hourly solar position (Altitude and Azimuth angles). The unsupervised time-series clustering framework successfully bypasses these operations by implicitly and automatically extracting these non-linear spatiotemporal relationships from the raw MRT data streams in less than 30 s. This computational efficiency and the specific choice of the algorithmic pipeline support a significant advancement over the current state-of-the-art. Recently, Nevers et al. (2025) demonstrated the validity of using time series clustering to isolate microclimatic zones at a neighbourhood scale. The authors suggest that their approach can be applied to any urban geometry by means of high-resolution modelling and extended to multi-day dataset.

By coupling our machine learning pipeline with the ultrafast LUCIDiT model, this data-mining approach can be scaled to the urban district level. Furthermore, the showed quantitative benchmarking proves higher technical clustering efficacy of raw K-means over shape-matching alternatives, like K-shape, within environmental

frameworks. The shift-invariant algorithms are valuable for a general pattern recognition, but they lose all the information concerning the solar radiation exposition. Solar radiation is fundamental to differentiate orientation-driven urban geometries (e.g., morning vs. afternoon thermal peaks).

Furthermore, by integrating the alpha-channel opacity modulation, based on absolute thermal threshold exceedance ($MRT > 40^{\circ}\text{C}$), the updated visualization framework directly addresses a limit of unsupervised classification schemes in climate studies, namely, the loss of absolute physical magnitudes. Standard clustering procedures mask cumulative heat stress by grouping pixels, based on relative trend variance, potentially conflating areas with identical dynamic shapes but different absolute temperature values. The resulting map (Fig. 8) acts as a dual-layer decision tool. The hue (colour) identifies the specific thermal morphology archetype, informing planners on what kind of intervention is required (e.g., high-canopy trees to block morning solar access in Cluster 2, or cool pavements to minimize delayed afternoon thermal storage in Cluster 1). The opacity (alpha) explicitly ranks the risk prioritization, flagging where the intervention is most urgently needed based on cumulative heat stress duration (Class D vs. Class A).

From an operational perspective, these findings enable targeted, boundary-focused mitigation strategies. Instead of designing logistically unfeasible, large-scale modifications in the centre of historical open squares (Cluster 0), planners can focus interventions along the transitional edges where Cluster 0 intersects with shaded archetypes. By means of a systematic extension of green canopies from these boundaries inward the consider cluster it is possible to carry out an economical optimization and an incremental reduction of the urban thermal footprint.

5.2 Limits and future outlook

While the machine learning clustering algorithm exhibits high mathematical robustness, computational scalability, and physical consistency, the proposed framework accuracy remains intrinsically bound to the boundary conditions and assumptions of the underlying microclimatic model used to generate the input dataset. Therefore, the limitations of this study concern useful data-accessibility and physical modelling-oriented rather than cluster algorithmic choice.

The domain adopts a binary classification for ground surface: with greenery (grass) and without greenery (asphalt-like). As a result, specific urban components such as water bodies are approximated as asphalt surfaces. While this represents a conservative, dark-surface "worst-case scenario" for radiative absorption and radiative heat exchange, it disregards the different

thermal capacity and local cooling effects of urban blue infrastructures.

Moreover, although cooling effect of vegetation is computed via the dynamic calculation of latent heat fluxes using the Penman–Monteith equation, the current framework is not coupled with a multi-layer hydrological soil model or explicit plant hydraulics. Consequently, tree transpiration rates do not adapt to localized hourly variations in soil moisture depletion, and complex biochemical processes like photosynthesis are excluded.

Finally, urban trees are simulated as a single macro-species with uniform thermophysical parameters (such as leaf transmissivity and stomatal resistance). Although these kinds of data are usually difficult to be collected, local variations in leaf area density or species-specific thermal behaviours are currently not captured within the simulation layout. Future research development will focus on coupling the machine learning framework with multi-physical microclimate digital twins to integrate dynamic soil–water–atmosphere interactions, enriching vegetation variability. Additionally, testing the transferability of the extracted cluster archetypes across different geographic latitudes and highly different architectural fabrics will further generalize the framework as a universal tool for climate-resilient urban design.

6 Conclusions

This study proposes a novel, highly scalable, data-driven framework to autonomously identify and map critical microclimatic areas in complex, real-world urban environments. By process a vast domain of 5.5 Km^2 (comprising over 4 million individual MRT time series), the analysis quantitatively demonstrated that applying the K-Means algorithm directly to raw, high-resolution data significantly outperforms both heuristic baselines and shape-matching techniques.

From a computational perspective, the raw K-Means pipeline exhibited efficiency, classifying the entire district on a laptop in under 30 s. From a physical perspective, the unsupervised algorithm autonomously reconstructed the thermophysical 3D urban morphology without any a priori knowledge of spatial coordinates, building footprints, or geometric boundaries. By strictly learning from the temporal variance and absolute amplitude of the data, K-Means avoided the rigid, fragmented "salt-and-pepper" boundaries typical of heuristic thresholds, and it preserved the crucial temporal synchronization of solar peaks that the shift-invariant K-Shape algorithm destroyed.

The physical robustness of the clustering was rigorously validated through a novel 2D Thermal Phase Space, proving that the four extracted thermal morphologies correspond to distinct, real-world thermodynamic and thermo-physical archetypes: open surfaces, west-facing

exposures, east-facing exposures, and shadow areas. This data-driven partitioning is perfectly corroborated by localized morphological View Factors (F_{sky} , F_{wall} , F_{veg}).

From an operational point of view, this methodology effectively links advanced physics-based microclimate simulations and actionable urban planning aimed at enhancing urban sustainability, resilience, and thermal comfort. By cross-referencing temporal clusters with an absolute thermal threshold exceedance analysis integrated via alpha-channel opacity modulation, the framework successfully retains both categorical trend types and absolute heat stress magnitudes. This translates computationally heavy MRT streams into an intuitive, dual-layer spatial diagnostic tool, enabling non-expert stakeholders and decision-makers to instantly identify, prioritize, and allocate targeted climate change mitigation measures.

Future developments of the method will leverage these spatial maps as the foundation for automated outdoor thermal comfort conditions optimization. Using the ultra-fast LUCIDiT model, targeted mitigation strategies—such as the progressive shading of the transitional boundaries around the most critical highly irradiated open spaces (Cluster 0)—will be simulated to dynamically evaluate how this microclimatic clusters evolve after specific intervention. While the optimal number of clusters ($k=4$) was explicitly constrained in this study to guarantee physical consistency with directional solar exposure, future research could explore dynamic, mathematically driven hyperparameter optimizations adapted for different climatic zones. Ultimately, integrating this unsupervised clustering framework with generative AI and automated optimization techniques, such as genetic algorithms, it will also enable the generative design of economically viable, highly localized, and physically robust urban resilience plans.

7 Nomenclature

CFD Computational Fluid Dynamic

DSM Digital Surface Model

k Number of clusters

LST Land Surface Temperature

ML Machine Learning

MRT Mean Radiant Temperature

OSM OpenStreetMaps

PET Physiological Environmental Temperature

SVF Sky View Factor

UHI Urban Heat Island

t_c temporal centroid

TWY Typical Weather Year

7.1 Greek symbols

$\Delta T_{max-min}$ Global Thermal range, °C

ΔT_{PM-AM} Thermal Asymmetry Index, °C

Institutional review board statement

Not applicable.

Informed consent statement

Not applicable.

Authors' contributions

M.B.: Writing, Methodology, Investigation, Validation, Software, Data curation, Conceptualization; C.B.: Writing, Review, Editing, Supervision, Funding Acquisition, Conceptualization; F.B.: Writing, Review and Editing. All authors have read and agreed to the published version of the manuscript.

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Data availability

The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Declarations

Competing interests

The authors declare no conflicts of interest and that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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